

Math ma'am Academy

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Std VI — Dr. Homi Bhabha Balvaidnyanik Competition

TOPICS: Plant Classification & Morphology · Animal Classification & Body Functions · Respiration · Matter & Changes of State · Force, Work & Energy · Units & Measurements · Separation of Substances · Metals & Non-metals · Types of Changes & Rusting · Waves & Sound · Nutrition · Water & Aquatic Plants · Levers · Diseases & Human Health

Time: 90 Minutes**Marks: 100****Questions: 100**

INSTRUCTIONS: 1. All questions are compulsory. Each question carries **1 MARK**; there is **NO NEGATIVE MARKING**. 2. Each question has **FOUR** alternatives — exactly **ONE** is correct. 3. Mark all answers on the **OMR SHEET** provided at the end by completely darkening the appropriate circle with a blue or black ball pen. 4. Questions are arranged by topic section; read each question carefully. 5. Rough work may be done on the blank space. 6. Calculators or electronic devices are **NOT PERMITTED**.

- A plant has a soft green stem, completes its life cycle within a short period and is generally small in height. It is most likely a:

A) Tree B) Herb C) Shrub D) Climber
- Which combination correctly represents a dicot plant?

A) Parallel venation + fibrous root + 1 cotyledon B) Reticulate venation + tap root + 2 cotyledons C) Parallel venation + tap root + 2 cotyledons
D) Reticulate venation + fibrous root + 1 cotyledon
- A plant has parallel venation but its roots are accidentally damaged. Which feature would provide the strongest additional evidence that it is a monocot?

A) Presence of flowers B) Fibrous root system C) Green leaves D) Presence of stem
- Which sequence represents increasing complexity of the plant groups?

A) Angiosperms -> Gymnosperms -> Pteridophytes -> Bryophytes B) Bryophytes -> Pteridophytes -> Gymnosperms -> Angiosperms
C) Pteridophytes -> Bryophytes -> Angiosperms -> Gymnosperms D) Bryophytes -> Gymnosperms -> Pteridophytes -> Angiosperms
- Which pair is incorrectly matched?

A) Bryophyta - Moss B) Pteridophyta - Fern C) Gymnosperm - Mango D) Angiosperm - Rose
- A plant possesses seeds but does not produce flowers or fruits. It is most likely a:

A) Bryophyte B) Pteridophyte C) Gymnosperm D) Angiosperm
- Which feature most clearly distinguishes angiosperms from gymnosperms?

A) Presence of roots B) Presence of stems C) Seeds enclosed within fruits D) Ability to perform photosynthesis
- Which of the following is correctly arranged?

A) Herb -> Neem, Shrub -> Rose, Tree -> Wheat B) Herb -> Wheat, Shrub -> Rose, Tree -> Neem C) Herb -> Mango, Shrub -> Grass, Tree -> Rose
D) Herb -> Rose, Shrub -> Wheat, Tree -> Grass
- A climber generally differs from a creeper because a climber:

A) Has no stem B) Grows only underground C) Uses support to grow upward D) Always has a woody stem
- Which combination contains only monocot characteristics?

A) Two cotyledons, tap root, reticulate venation B) One cotyledon, fibrous root, parallel venation C) Two cotyledons, fibrous root, parallel venation
D) One cotyledon, tap root, reticulate venation
- Which part of a flower receives pollen grains during pollination?

A) Anther B) Filament C) Stigma D) Ovary
- Which sequence correctly represents the male reproductive part of a flower?

A) Stigma -> Style -> Ovary B) Anther -> Filament C) Ovary -> Ovule -> Stigma D) Sepal -> Petal -> Anther
- If the anthers of a flower are removed before maturity, which process is directly affected?

A) Formation of ovules B) Production of pollen grains C) Formation of sepals D) Absorption of water
- Which statement about annual, biennial and perennial plants is correct?

A) Annual plants live for exactly two years B) Biennial plants complete their life cycle in one year C) Perennial plants can live for several years
D) All perennial plants are trees
- Which characteristic is common to all vertebrates?

A) They have lungs B) They have a backbone C) They lay eggs D) They are warm-blooded
- Which pair contains only warm-blooded animals?

A) Frog and lizard B) Fish and snake C) Birds and mammals D) Crocodile and fish
- Which pair contains animals that are both generally oviparous?

A) Cow and dog B) Hen and frog C) Human and whale D) Cat and rabbit

- D) Viviparous animals never have eggs**
-
19. Which of the following is an example of a cold-blooded vertebrate?
A) Sparrow B) Human C) Lizard D) Cow
-
20. Which statement best describes external respiration?
A) Breakdown of glucose inside cells B) Exchange of gases between respiratory surface and surroundings/blood
C) Production of ATP inside mitochondria only D) Movement of food through the alimentary canal
-
21. During aerobic respiration, glucose is broken down mainly in the presence of:
A) Nitrogen B) Carbon dioxide C) Oxygen D) Hydrogen
-
22. Which is a correct comparison?
A) Aerobic respiration occurs without oxygen and always produces alcohol B) Anaerobic respiration requires more oxygen than aerobic respiration
C) Aerobic respiration generally releases more energy than anaerobic respiration D) Both always produce exactly the same products
-
23. A muscle cell temporarily does not receive sufficient oxygen during strenuous activity. Which process can increase?
A) Photosynthesis B) Anaerobic respiration C) Transpiration D) Digestion
-
24. Which gas is mainly taken in during aerobic respiration?
A) Oxygen B) Nitrogen C) Carbon monoxide D) Hydrogen
-
25. According to the particle theory of matter, increasing the temperature of a substance generally causes its particles to:
A) Stop moving B) Lose all kinetic energy C) Gain kinetic energy D) Become completely weightless
-
26. Which change involves absorption of heat?
A) Freezing B) Condensation C) Melting D) Deposition
-
27. A wet cloth dries faster on a hot, windy day mainly because:
A) Water changes directly into ice B) Evaporation becomes faster C) Condensation becomes faster D) Water particles stop moving
-
28. Which pair represents opposite changes of state?
A) Melting and freezing B) Melting and evaporation C) Condensation and evaporation D) Both A and C
-
29. When water vapour changes into liquid water, the process is:
A) Sublimation B) Evaporation C) Condensation D) Melting
-
30. A substance changes directly from solid to gas without becoming liquid. This process is:
A) Freezing B) Sublimation C) Condensation D) Vaporisation
-
31. Which observation provides the strongest evidence that particles of matter are continuously moving?
A) A stone has fixed shape B) Perfume smell spreads throughout a room C) A metal spoon has mass D) Water occupies space
-
32. A ball thrown vertically upward continues to move upward for some time after leaving the hand. During this upward motion, the main force acting on it is:
A) Muscular force upward B) Gravitational force downward C) Frictional force upward D) Magnetic force downward
-
33. Work is said to be done on an object when:
A) Force acts but there is no displacement B) Displacement occurs in a direction having a component along the force C) The object has mass
D) The object is stationary
-
34. Which situation involves no mechanical work on the object, despite a force being applied?
A) Pushing a box and moving it forward B) Lifting a bag upward C) Holding a heavy bag stationary D) Kicking a football
-
35. A 2 kg object and a 5 kg object move with the same velocity. Which has greater kinetic energy?
A) 2 kg object B) 5 kg object C) Both have equal kinetic energy D) Cannot be determined
-
36. If the velocity of an object is doubled while its mass remains constant, its kinetic energy becomes:
A) Half B) Double C) Four times D) Eight times
-
37. Which object has the greatest gravitational potential energy, assuming all are on Earth?
A) 2 kg object at 1 m B) 1 kg object at 4 m C) 4 kg object at 1 m D) 1 kg object at 2 m
-
38. A stretched rubber band stores mainly:
A) Chemical energy B) Elastic potential energy C) Sound energy D) Nuclear energy
-
39. Which conversion mainly occurs in an electric fan?
A) Electrical -> mechanical B) Mechanical -> electrical C) Chemical -> light D) Heat -> chemical
-
40. A falling object changes its energy mainly from:
A) Kinetic to potential B) Potential to kinetic C) Chemical to electrical D) Sound to light
-

- D)** An object with no mass
-
42. Which is the correct SI unit of length?
A) Centimetre **B)** Metre **C)** Kilometre **D)** Millimetre
-
43. A student measures the length of a pencil as 12.5 cm. This is equal to:
A) 1.25 m **B)** 0.125 m **C)** 125 m **D)** 0.0125 m
-
44. Which instrument is most appropriate for measuring the temperature of a liquid in a school experiment?
A) Measuring cylinder **B)** Thermometer **C)** Spring balance **D)** Stopwatch
-
45. Which measurement is most likely to require a stopwatch?
A) Mass of an object **B)** Temperature of water **C)** Time taken by a pendulum to complete oscillations **D)** Volume of milk
-
46. A student wants to measure 35 mL of water accurately. Which instrument is most appropriate?
A) Measuring cylinder **B)** Metre scale **C)** Thermometer **D)** Balance
-
47. Which statement is correct?
A) 1 m = 1000 cm **B)** 1 km = 1000 m **C)** 1 cm = 1000 mm **D)** 1 L = 1000 kg
-
48. A temperature of 0 °C corresponds approximately to:
A) 0 K **B)** 100 K **C)** 273 K **D)** 373 K
-
49. A mixture contains iron filings and sand. The most suitable first method of separation is:
A) Filtration **B)** Magnetic separation **C)** Evaporation **D)** Distillation
-
50. Which method is best suited to separate insoluble solid particles from a liquid?
A) Filtration **B)** Sublimation **C)** Separating funnel **D)** Magnetic separation
-
51. A mixture contains common salt and sand. Which sequence is most appropriate to obtain dry salt?
A) Filtration → add water → evaporation **B)** Add water → filtration → evaporation **C)** Evaporation → filtration → add water
D) Magnetic separation → evaporation
-
52. Which technique is used to separate different coloured pigments present in ink?
A) Centrifugation **B)** Paper chromatography **C)** Winnowing **D)** Sedimentation
-
53. Cream can be separated from milk mainly by:
A) Filtration **B)** Centrifugation **C)** Sieving **D)** Sublimation
-
54. A mixture of two immiscible liquids such as oil and water can best be separated using:
A) Separating funnel **B)** Filter paper **C)** Evaporating dish **D)** Magnet
-
55. Which method is most suitable for obtaining pure water from salt solution?
A) Filtration **B)** Distillation **C)** Sieving **D)** Handpicking
-
56. The separation of husk from heavier grains using moving air is:
A) Threshing **B)** Winnowing **C)** Filtration **D)** Crystallisation
-
57. Which method is used when the components of a mixture have significantly different particle sizes?
A) Sieving **B)** Distillation **C)** Separating funnel **D)** Chromatography
-
58. Photosynthetic pigments from a leaf can be separated using:
A) Paper chromatography **B)** Sedimentation **C)** Magnetic separation **D)** Handpicking
-
59. Which of the following is a liquid metal at room temperature?
A) Iron **B)** Mercury **C)** Sodium **D)** Aluminium
-
60. Which is a liquid non-metal at room temperature?
A) Sulphur **B)** Bromine **C)** Carbon **D)** Phosphorus
-
61. Which pair contains only metals?
A) Iron and sulphur **B)** Copper and aluminium **C)** Carbon and oxygen **D)** Phosphorus and chlorine
-
62. An alloy is best described as:
A) A pure metal only **B)** A mixture of two or more metals or a metal with another element **C)** A non-metallic compound only
D) A naturally occurring mineral only
-
63. Which property is generally associated with metals?
A) Brittleness **B)** Malleability **C)** Poor electrical conductivity **D)** Dull appearance
-

65. Diamond and graphite are:
A) Compounds of different metals B) Allotropes of carbon C) Isotopes of carbon D) Mixtures containing carbon
66. Which element is generally considered a non-metal?
A) Magnesium B) Calcium C) Sulphur D) Sodium
67. Which metal is placed above copper in the usual reactivity series?
A) Silver B) Gold C) Iron D) Platinum
68. Which sequence is correctly arranged from more reactive to less reactive?
A) $K > Na > Ca > Mg$ B) $Mg > Ca > Na > K$ C) $Na > K > Mg > Ca$ D) $Ca > K > Mg > Na$
69. Which pair represents an important use of metals based on their properties?
A) Copper - electrical wiring B) Sulphur - electrical wiring C) Phosphorus - making cooking vessels D) Oxygen - making metal wires
70. Which is an example of a chemical change?
A) Melting ice B) Cutting paper C) Rusting of iron D) Dissolving sugar in water
71. Which observation most strongly indicates that a chemical change has occurred?
A) Change in shape only B) Change in state only C) Formation of a new substance D) Breaking an object into pieces
72. Rusting of iron requires mainly:
A) Oxygen only B) Water only C) Oxygen and moisture D) Nitrogen and sunlight
73. Which method protects iron from rusting by providing a protective zinc coating?
A) Galvanisation B) Distillation C) Crystallisation D) Winnowing
74. Painting an iron object prevents rusting mainly because paint:
A) Increases the iron content B) Prevents contact with air and moisture C) Converts iron into a non-metal
D) Removes oxygen from iron permanently
75. Which condition would generally make iron rust faster?
A) Dry air B) Complete absence of oxygen C) Moist air containing oxygen D) Coating with oil
76. In a transverse wave, particles of the medium vibrate:
A) Parallel to the direction of wave propagation B) Perpendicular to the direction of wave propagation C) Only upward D) Only downward
77. In a longitudinal wave, particles vibrate:
A) Perpendicular to wave motion B) Parallel to wave motion C) In circular paths only D) Without transferring energy
78. Sound travelling through air is an example of:
A) Transverse mechanical wave B) Longitudinal mechanical wave C) Electromagnetic transverse wave D) Static wave
79. Which pair correctly represents a longitudinal wave feature?
A) Crest and trough B) Compression and rarefaction C) Peak and valley only D) Node and antinode only
80. A child has poor vision in dim light. Deficiency of which vitamin is most likely involved?
A) Vitamin A B) Vitamin C C) Vitamin D D) Vitamin K
81. Which deficiency is associated with scurvy?
A) Vitamin A B) Vitamin B C) Vitamin C D) Vitamin D
82. Rickets is mainly associated with deficiency of:
A) Vitamin A B) Vitamin C C) Vitamin D D) Vitamin B12
83. A person suffering from anaemia may have a deficiency of:
A) Iron B) Iodine C) Calcium D) Sodium
84. Goitre is commonly associated with deficiency of:
A) Iron B) Iodine C) Calcium D) Potassium
85. Which pair is incorrectly matched?
A) Vitamin C - Scurvy B) Vitamin D - Rickets C) Iodine - Goitre D) Iron - Night blindness
86. Which combination would be most useful in preventing iron-deficiency anaemia?
A) Iron-rich foods B) Iodine-rich foods only C) Vitamin D alone D) Calcium supplements alone
87. Approximately what percentage of Earth's water is present in oceans and seas?
A) 50% B) 70% C) 97% D) 25%

89. Which reservoir contains the largest proportion of Earth's freshwater?
A) Rivers B) Atmosphere C) Glaciers and ice caps D) Living organisms
90. Water hyacinth is scientifically known as:
A) Eichhornia crassipes B) Leishmania donovani C) Plasmodium vivax D) Bryophyllum pinnatum
91. Water hyacinth is considered harmful in many water bodies mainly because it:
A) Increases sunlight penetration B) Rapidly covers the surface and reduces available oxygen C) Converts freshwater into seawater
D) Prevents all evaporation
92. The rapid spread of water hyacinth is mainly due to:
A) Slow seed formation only B) Rapid vegetative propagation C) Absence of leaves D) Underground flowering
93. In a first-class lever:
A) Load is always between fulcrum and effort B) Effort is always between fulcrum and load C) Fulcrum lies between effort and load
D) Fulcrum is absent
94. Which is an example of a second-class lever?
A) Scissors B) Wheelbarrow C) Tweezers D) Fishing rod
95. Which arrangement represents a third-class lever?
A) Fulcrum - Load - Effort B) Fulcrum - Effort - Load C) Load - Fulcrum - Effort D) Effort - Load - Fulcrum
96. A pair of scissors is generally classified as:
A) First-class lever B) Second-class lever C) Third-class lever D) Not a lever
97. Which of the following is a congenital disease/condition?
A) Cholera B) Haemophilia C) Malaria D) Typhoid
98. Which statement is correct?
A) Ringworm is caused by a worm B) Malaria is caused by a mosquito C) Malaria is caused by Plasmodium and transmitted by a mosquito
D) Kala-azar is caused by a bacterium
99. Which pair is correctly matched?
A) Kala-azar - Leishmania B) Taeniasis - Roundworm C) Ascariasis - Tapeworm D) Scabies - Bacterium
100. Consider the following statements: 1. Bacteria are prokaryotic organisms. 2. Protozoans are generally unicellular eukaryotes. 3. Ringworm is caused by a fungus. 4. A vector is always the causative organism of a disease. Which option is correct?
A) Only 1 and 2 B) Only 1, 2 and 3 C) Only 2, 3 and 4 D) All four

———— END OF QUESTION PAPER ————

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Scan to attempt
online**O M R A N S W E R S H E E T**

Student Name : _____ Std : _____

Exam Name : _____ Roll No. : _____

Batch / Division : _____ Date : _____ Marks Obtained : _____

How to mark: Completely darken **one** circle per question using a blue or black ball pen. Do not use pencil. Overwriting, ticks, crosses or more than one darkened circle will be treated as a wrong answer.

Example: ☐ ☐

1	☐ A ☐ B ☐ C ☐ D	26	☐ A ☐ B ☐ C ☐ D	51	☐ A ☐ B ☐ C ☐ D	76	☐ A ☐ B ☐ C ☐ D
2	☐ A ☐ B ☐ C ☐ D	27	☐ A ☐ B ☐ C ☐ D	52	☐ A ☐ B ☐ C ☐ D	77	☐ A ☐ B ☐ C ☐ D
3	☐ A ☐ B ☐ C ☐ D	28	☐ A ☐ B ☐ C ☐ D	53	☐ A ☐ B ☐ C ☐ D	78	☐ A ☐ B ☐ C ☐ D
4	☐ A ☐ B ☐ C ☐ D	29	☐ A ☐ B ☐ C ☐ D	54	☐ A ☐ B ☐ C ☐ D	79	☐ A ☐ B ☐ C ☐ D
5	☐ A ☐ B ☐ C ☐ D	30	☐ A ☐ B ☐ C ☐ D	55	☐ A ☐ B ☐ C ☐ D	80	☐ A ☐ B ☐ C ☐ D
6	☐ A ☐ B ☐ C ☐ D	31	☐ A ☐ B ☐ C ☐ D	56	☐ A ☐ B ☐ C ☐ D	81	☐ A ☐ B ☐ C ☐ D
7	☐ A ☐ B ☐ C ☐ D	32	☐ A ☐ B ☐ C ☐ D	57	☐ A ☐ B ☐ C ☐ D	82	☐ A ☐ B ☐ C ☐ D
8	☐ A ☐ B ☐ C ☐ D	33	☐ A ☐ B ☐ C ☐ D	58	☐ A ☐ B ☐ C ☐ D	83	☐ A ☐ B ☐ C ☐ D
9	☐ A ☐ B ☐ C ☐ D	34	☐ A ☐ B ☐ C ☐ D	59	☐ A ☐ B ☐ C ☐ D	84	☐ A ☐ B ☐ C ☐ D
10	☐ A ☐ B ☐ C ☐ D	35	☐ A ☐ B ☐ C ☐ D	60	☐ A ☐ B ☐ C ☐ D	85	☐ A ☐ B ☐ C ☐ D
11	☐ A ☐ B ☐ C ☐ D	36	☐ A ☐ B ☐ C ☐ D	61	☐ A ☐ B ☐ C ☐ D	86	☐ A ☐ B ☐ C ☐ D
12	☐ A ☐ B ☐ C ☐ D	37	☐ A ☐ B ☐ C ☐ D	62	☐ A ☐ B ☐ C ☐ D	87	☐ A ☐ B ☐ C ☐ D
13	☐ A ☐ B ☐ C ☐ D	38	☐ A ☐ B ☐ C ☐ D	63	☐ A ☐ B ☐ C ☐ D	88	☐ A ☐ B ☐ C ☐ D
14	☐ A ☐ B ☐ C ☐ D	39	☐ A ☐ B ☐ C ☐ D	64	☐ A ☐ B ☐ C ☐ D	89	☐ A ☐ B ☐ C ☐ D
15	☐ A ☐ B ☐ C ☐ D	40	☐ A ☐ B ☐ C ☐ D	65	☐ A ☐ B ☐ C ☐ D	90	☐ A ☐ B ☐ C ☐ D
16	☐ A ☐ B ☐ C ☐ D	41	☐ A ☐ B ☐ C ☐ D	66	☐ A ☐ B ☐ C ☐ D	91	☐ A ☐ B ☐ C ☐ D
17	☐ A ☐ B ☐ C ☐ D	42	☐ A ☐ B ☐ C ☐ D	67	☐ A ☐ B ☐ C ☐ D	92	☐ A ☐ B ☐ C ☐ D
18	☐ A ☐ B ☐ C ☐ D	43	☐ A ☐ B ☐ C ☐ D	68	☐ A ☐ B ☐ C ☐ D	93	☐ A ☐ B ☐ C ☐ D
19	☐ A ☐ B ☐ C ☐ D	44	☐ A ☐ B ☐ C ☐ D	69	☐ A ☐ B ☐ C ☐ D	94	☐ A ☐ B ☐ C ☐ D
20	☐ A ☐ B ☐ C ☐ D	45	☐ A ☐ B ☐ C ☐ D	70	☐ A ☐ B ☐ C ☐ D	95	☐ A ☐ B ☐ C ☐ D
21	☐ A ☐ B ☐ C ☐ D	46	☐ A ☐ B ☐ C ☐ D	71	☐ A ☐ B ☐ C ☐ D	96	☐ A ☐ B ☐ C ☐ D
22	☐ A ☐ B ☐ C ☐ D	47	☐ A ☐ B ☐ C ☐ D	72	☐ A ☐ B ☐ C ☐ D	97	☐ A ☐ B ☐ C ☐ D
23	☐ A ☐ B ☐ C ☐ D	48	☐ A ☐ B ☐ C ☐ D	73	☐ A ☐ B ☐ C ☐ D	98	☐ A ☐ B ☐ C ☐ D
24	☐ A ☐ B ☐ C ☐ D	49	☐ A ☐ B ☐ C ☐ D	74	☐ A ☐ B ☐ C ☐ D	99	☐ A ☐ B ☐ C ☐ D
25	☐ A ☐ B ☐ C ☐ D	50	☐ A ☐ B ☐ C ☐ D	75	☐ A ☐ B ☐ C ☐ D	100	☐ A ☐ B ☐ C ☐ D

Student's Signature

Invigilator's / Examiner's Signature